

Appendix 1

a_1 as function of detector width δ_1

Finding a_1^{α} using ordinary integration

$$A = \left(\frac{\delta^2}{2\pi} \right)^{1/4} * \frac{\text{Exp}[-k_0^2 * \delta^2]}{\sqrt{\delta^2 + \frac{i*t}{2}}} * \left(\frac{\delta^2}{2\pi} \right)^{1/4} * \frac{\text{Exp}[-k_0^2 * \delta^2]}{\sqrt{\delta^2 - \frac{i*t}{2}}};$$

$$\text{Integrate}[\text{Exp}[(4 T * y * \delta^2 * k_0 - 2 y^2 * \delta^2) / (4 \delta^4 + T^2)], \{y, x_1 - \delta_1 / 2 - x_0, x_1 + \delta_1 / 2 - x_0\}];$$

$$\text{Const} = \text{Simplify}\left[\frac{e^{-2 \delta^2 k_0^2} \sqrt{\delta^2}}{\sqrt{2\pi} \sqrt{-\frac{i*t}{2} + \delta^2} \sqrt{\frac{i*t}{2} + \delta^2}} * \text{Exp}[(8 \delta^6 k_0^2) / (4 \delta^4 + T^2)]\right];$$

$$C_1[T_-, D_-] = \frac{1}{2} \left(-\text{Erf}\left[\frac{\sqrt{2} \delta (T k_0 + X_0 - X_d - \frac{D}{2})}{\sqrt{T^2 + 4 \delta^4}}\right] + \text{Erf}\left[\frac{\sqrt{2} \delta (T k_0 + X_0 - X_d + D / 2)}{\sqrt{T^2 + 4 \delta^4}}\right] \right);$$

Now lets compare

$$\psi[0, x_-, t_-] := \left(\frac{\delta^2}{2\pi} \right)^{1/4} * \frac{\text{Exp}[-k_0^2 * \delta^2]}{\sqrt{\delta^2 + \frac{i*t}{2}}} * \text{Exp}\left[\frac{(2 * \delta^2 * k_0 + i * (x - x_0))^2}{4 * \delta^2 + 2 * i * t}\right];$$

$$a_1[D_-] := \text{Abs}[\psi[0, X_d, \text{tm}[1]]] * \text{Sqrt}[D];$$

```

k0 = 20;
Xd = -3;
X0 = -5;
δ = 0.5;
Ti = -0.02;
Tf = 0.22;
dt = (Tf - Ti) / (n - 1);
n = 100;
tm[0] = Ti - dt;
tm[1] = Ti;
tm[n + 1] = Tf + dt;

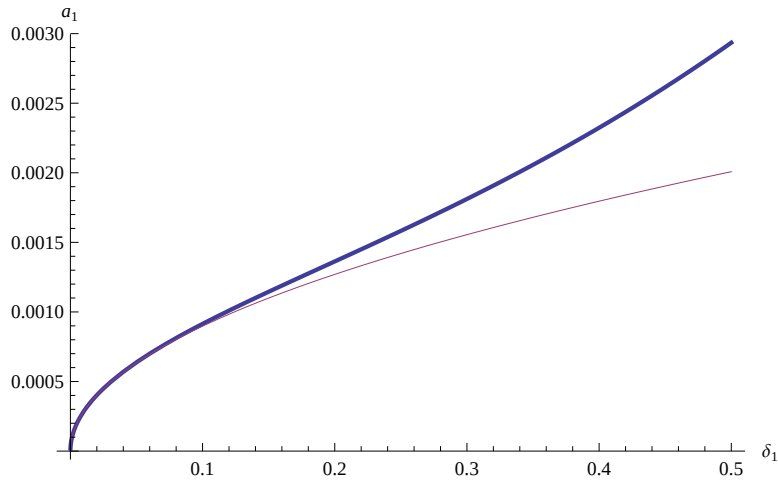
```

Finding a_1^{β} using small detector approximation

```

A = Plot[{Sqrt[C1[tm[1], δ1]], a1[δ1]}, {δ1, 0.00005, 0.5},
  PlotStyle -> {Thick, Thin}, AxesLabel -> {"δ1", "a1"}]

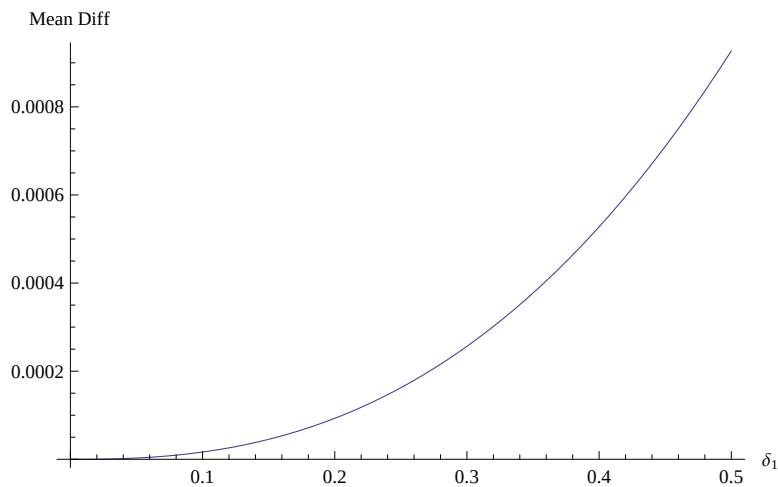
```



```

Diff1 = Plot[Abs[{Sqrt[C1[tm[1], δ1]] - a1[δ1]}],
  {δ1, 0.00005, 0.5}, AxesLabel -> {"δ1", "Mean Diff"}]

```



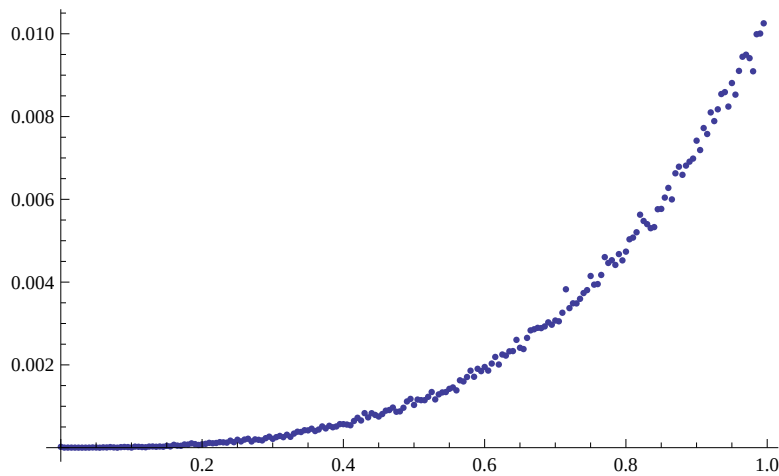
Finding a^γ_1 using random position approximation

```
m = 10 000;
```

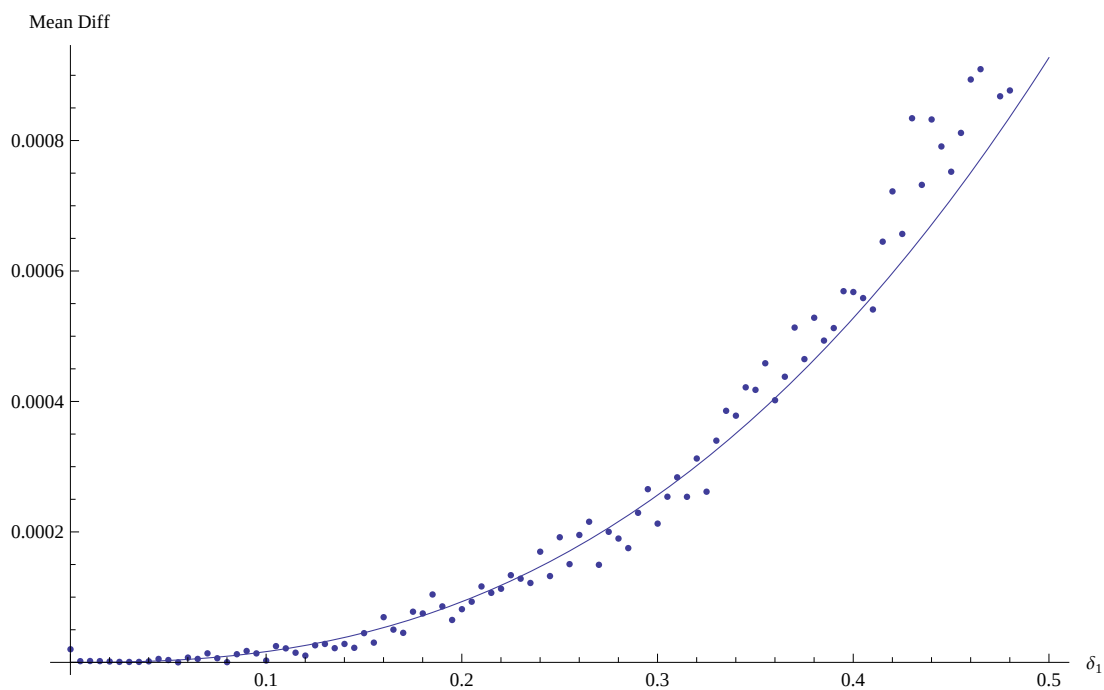
```
AverageList = {}; DiffList = {};
```

```
Do[{a = 0, i = 0; Do[a = a + Abs[ψ[0, RandomReal[{Xd - D1, Xd + D1}], tm[1]]] * Sqrt[D1],  
  {i, 1, 10 000 * D1}], AverageList = Append[AverageList, {D1, (a / (10 000 * D1))}],  
  DiffList = Append[DiffList, {D1, Abs[(√C1[tm[1], D1] - a / (10 000 * D1))]}],  
  {D1, .00005, 1, .005}]
```

```
Diff2 = ListPlot[DiffList]
```



```
Show[Diff1, Diff2]
```

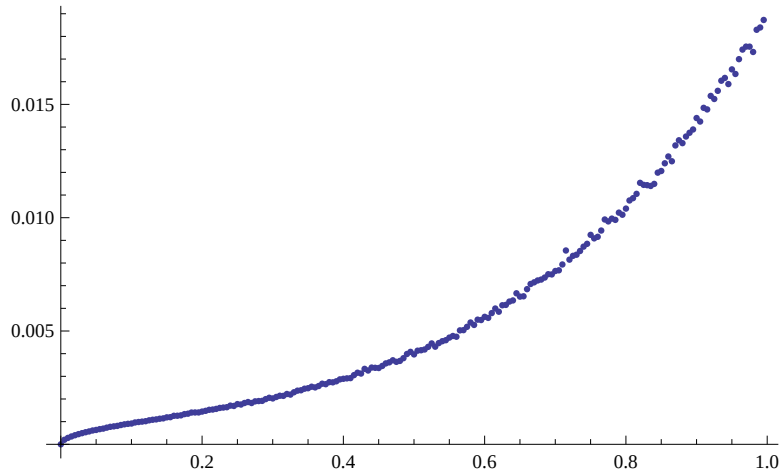


```

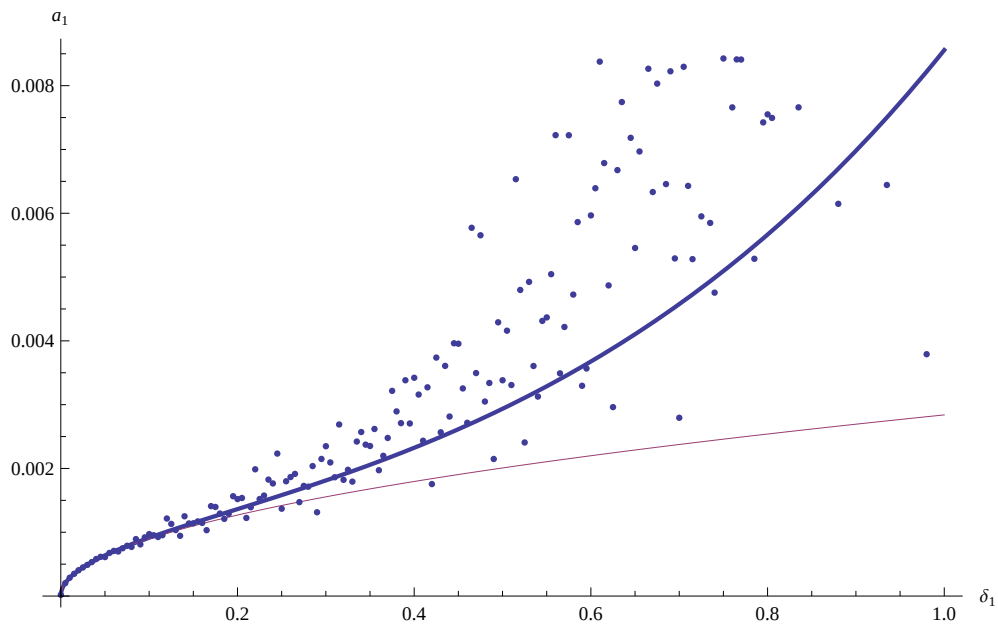
AverageList2 = {}; Do[{a = 0, i = 0;
  Do[a = a + Abs[ψ[0, RandomReal[{Xd - D1, Xd + D1}], tm[1]]] * Sqrt[D1], {i, 1, 20}],
  AverageList2 = Append[AverageList2, {D1, (a / (20))}], {D1, .00005, 1, .005}]

```

```
B = ListPlot[AverageList]
```



```
Show[A, c]
```



Show[A, B]

